SEDATION & ANALGESIA INITIATION

CAUTIONS TO USE OF THIS FLOWCHART:

MUSCLE RELAXATION (cannot use COMFORT-B))
OVERWEIGHT FOR AGE (see specific drug dosing information for ideal body weight for each drug)
STATUS EPILEPTICUS OR STATUS DYSTONICUS(sedation targets may be deeper)

EXCLUSIONS TO THIS FLOWCHART:

BRAINSTEM DEATH

FOR EACH PATIENT, PAY ATTENTION TO:

Age & Weight (if applicable, Ideal Body Weight)
Allergies & previous Medication History
Admission Reason
Anticipated Length of Stay
Infant/Child Developmental Stage & their individual behavioural "norms" (ask parent/carer)
Non-pharmacological comforters (toys, dummy, blanket, white noise, music, iPad, etc)
Existing Neurodisability
Baseline renal function
Baseline hepatic function

DOCUMENT:
Baseline COMFORT B & NISS Scores (SEDATION)
Baseline FACES or NRS score (PAIN)
Sedative & analgesic drugs already given in past 1hr (e.g. in theatre or ED)

Ventilated patient?

Likely duration of ventilation?

>48hrs

SHOCK OR HAEMODYNAMIC INSTABILITY +/- HEPATIC IMPAIRMENT?

Start FENTANYL

Consider LOADING dose 0.5-1micrograms/kg IBW

then INFUSION 1-4microgram/kg/hr

AFTER 40 MINUTES Reassess:

1) COMFORT B and/or NISS (Sedation scores)
2) NRS or FACES (Pain scores)

(See Scoring Tools)

If scores suggest SEDATION is an issue Start low-dose MIDAZOLAM INFUSION 0.03milligrams/kg/hr (i.e. 30micrograms/kg/hr)

If CVS too unstable for Midazolam, (or already started in theatre) use DEXMEDETOMIDINE INFUSION at 0.2 - 1.2 micrograms/kg/hr

If scores suggest PAIN is an issue, increment FENTANYL INFUSION in stages (MAX RATE 6micrograms/kg/hr)

TITRATE INFUSIONS TO TARGET SCORES See Dose Ceilings on Drug Info Quick Reference Guide

REPEAT SCORES 3-4 HOURLY FOR FIRST 12 HOURS OR UNTIL STEADY STATE

ANY OF:
Shock (lactate, ScvO2)?
Unstable CVS?
Liver impairment?
AKI?

NONE

AKI?

Start ALFENTANIL INFUSION 10-30microgram/kg/hr

If CVS stable, consider LOADING dose 50 micrograms/kg IBW over 10 MINUTES

AFTER 40 MINUTES Reassess:

1) COMFORT B and/or NISS (Sedation scores)
2) NRS or FACES (Pain scores)

(See Scoring Tools)

If scores suggest SEDATION is an issue Start low-dose MIDAZOLAM INFUSION 0.03milligrams/kg/hr (i.e. 30micrograms/kg/hr)

If CVS too unstable for Midazolam, (or already started in theatre) use DEXMEDETOMIDINE INFUSION 0.2 - 1.2 micrograms/kg/hr

If scores suggest PAIN is an issue, increment ALFENTANIL INFUSION in stages (MAX RATE 120micrograms/kg/hr)

TITRATE INFUSIONS TO TARGET SCORES See Dose Ceilings on Drug Info Quick Reference Guide

REPEAT SCORES 3-4 HOURLY FOR FIRST 12 HOURS OR UNTIL STEADY STATE

<48hrs

AGE/ IDEAL BODYWEIGHT?

<50kg

Start MORPHINE: LOADING dose 50micrograms/kg IBW as bolus then INFUSION 20microgram/kg/hr

Then start CLONIDINE INFUSION 1microgram/kg/hr

If required, use slow IV bolus of Clonidine (0.5-1mcg/kg over 30 min) to achieve plane of sedation

Or PROPOFOL very short infusion (<48hrs) may be used instead. CONSULTANT DECISION

(see ALERT BOX)

AFTER 40 MINUTES Reassess:

1) COMFORT B and/or NISS (Sedation scores)
2) NRS or FACES (Pain scores)

(See Scoring Tools)

TITRATE INFUSIONS TO TARGET SCORES

Consider addition of MIDAZOLAM INFUSION 0.1milligrams/kg/hr (100micrograms/kg/hr)

If scores suggest PAIN is an issue, increment MORPHINE INFUSION in stages (MAX RATE 50micrograms/kg/hr)

TITRATE INFUSIONS TO TARGET SCORES See Dose Ceilings on Drug Info Quick Reference Guide

REPEAT SCORES 3-4 HOURLY FOR FIRST 12 HOURS OR UNTIL STEADY STATE

>12years AND >50kg

Start MORPHINE: LOADING dose 50micrograms/kg IBW as bolus then INFUSION 20microgram/kg/hr

Then start PROPOFOL INFUSION at 1mg/kg/hr

AFTER 40 MINUTES

Reassess:
1) COMFORT B and/or NISS (Sedation scores)
2) NRS or FACES (Pain scores)

(See Scoring Tools)

TITRATE INFUSIONS TO TARGET SCORES

Consider addition of Clonidine INFUSION at 1mcg/kg/hr

TITRATE INFUSIONS TO TARGET SCORES See Dose Ceilings on Drug Info Quick Reference Guide

REPEAT SCORES 3-4 HOURLY FOR FIRST 12 HOURS OR UNTIL STEADY STATE

COMFORT B score ranges:

>17 undersedated OR sore (use pain score to decide)

12-17 ideal range most patients

10-12 increased cough suppression

<10 neuroprotection and/or induced coma for control of oxygen demand or other reason

Titrating COMFORT B scores:

If score too low or too high (and patient is not in PAIN) adjust SEDATIVE infusion up or down by 25% and reassess score after 1 hour.

Remember opiates also contribute to sedation in small infants (<6m)

CAUTION:

DEXMEDETOMIDINE may affect body temperature control, particularly in overweight patients. Monitor temperature hourly or continuously. STOP if **T>38.5°C for 2 consecutive hours**

ALERT:

PROPOFOL INFUSION SYNDROME (PrIS) **IS FATAL**

PROPOFOL **MUST NOT** BE USED FOR LONGER THAN **48 HRS** AT A TIME OR AT DOSES **>4mg/kg/hr**

ANY DEVIATION FROM THIS RULE IS A **CONSULTANT DECISION** AND MUST BE **CLEARLY DOCUMENTED** IN THE SEDATION PLAN.

PROLONGED PROPOFOL USE (>48HOURS) **MUST** BE ACCOMPANIED BY:
6 HOURLY LACTATE
DAILY TRIGLYCERIDE & LFTs and DAILY PAUSES

STOP PROPOFOL IMMEDIATELY IF ANY SIGN OF Pr.I.S. & CALL CONSULTANT

DAILY PAUSE METHOD:
Pause propofol infusion until patient becomes uncomfortable (COMFORT-B score >17) then give an age-appropriate bolus and restart the infusion at 25% less than previously.